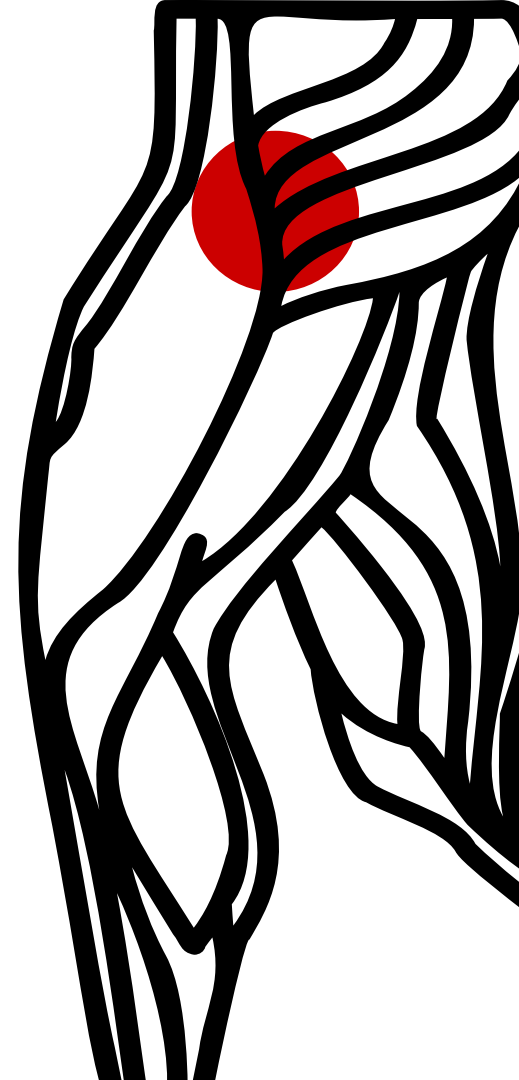


BIOMECHANICS:

The study of the movement of a living body; how muscles, bones, tendons, and ligaments work together to produce movement.



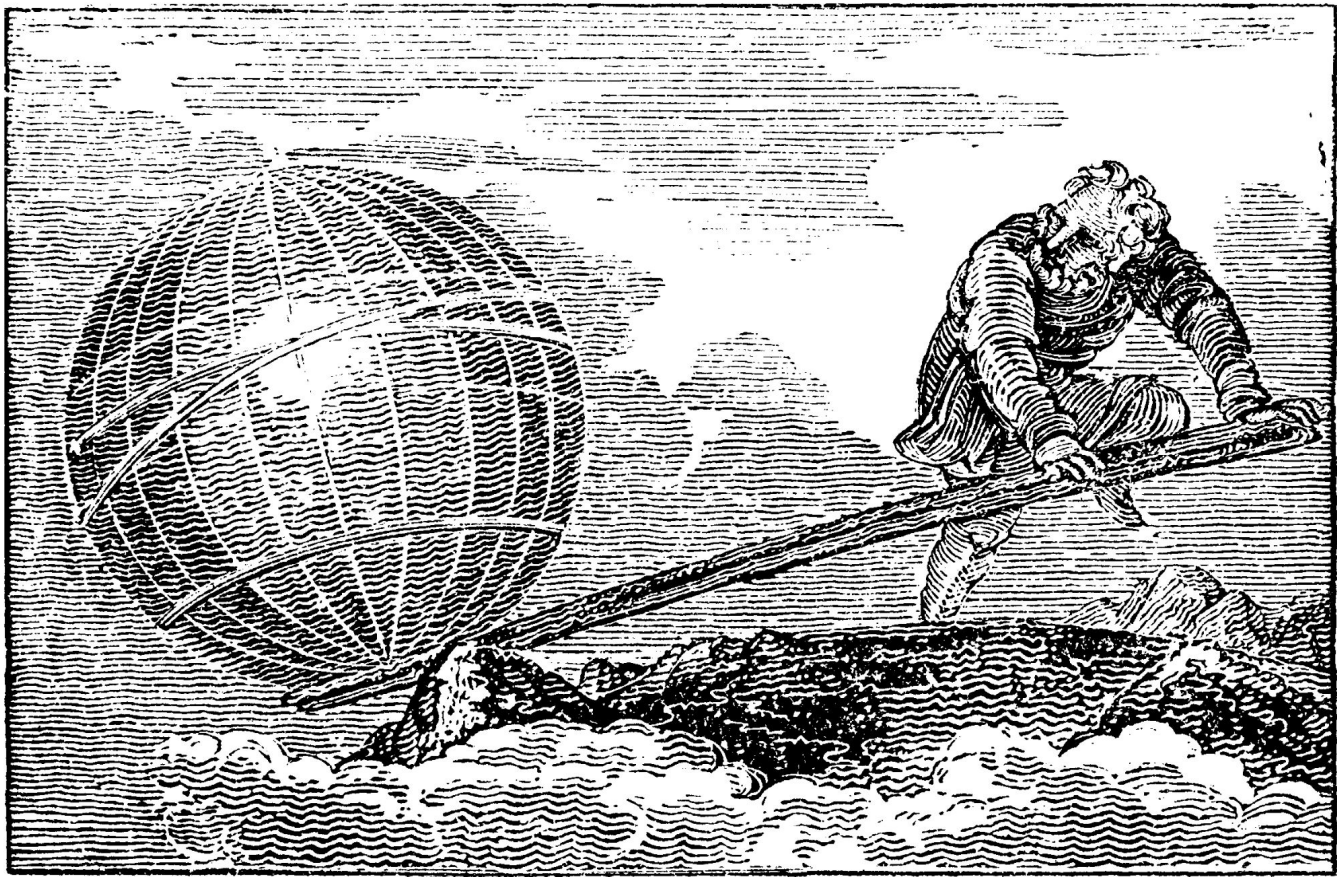


Photo courtesy of Wikimedia Commons

"Give me a lever long enough, and fulcrum on which to place it, and I shall move the world."
-Archimedes

TABLE OF CONTENTS

01

LEVERS

What is a lever and what are its components?

02

MECHANICAL ADVANTAGE

How do levers enhance movement?

03

BIOMECHANICS

What levers exist in the body and how do they enhance movement?

04

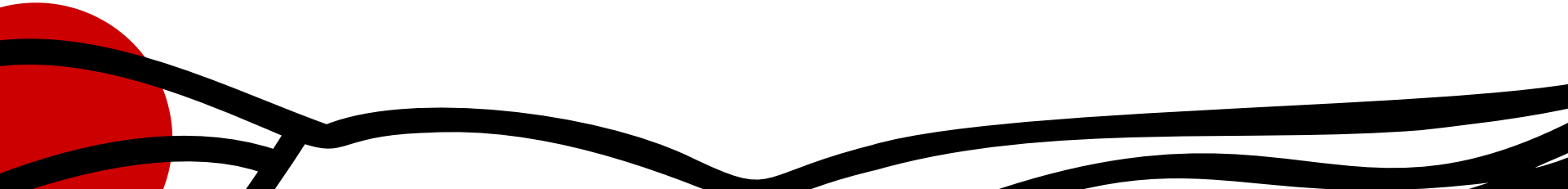
TUTORIAL

Measuring mechanical advantage on MorphoSource



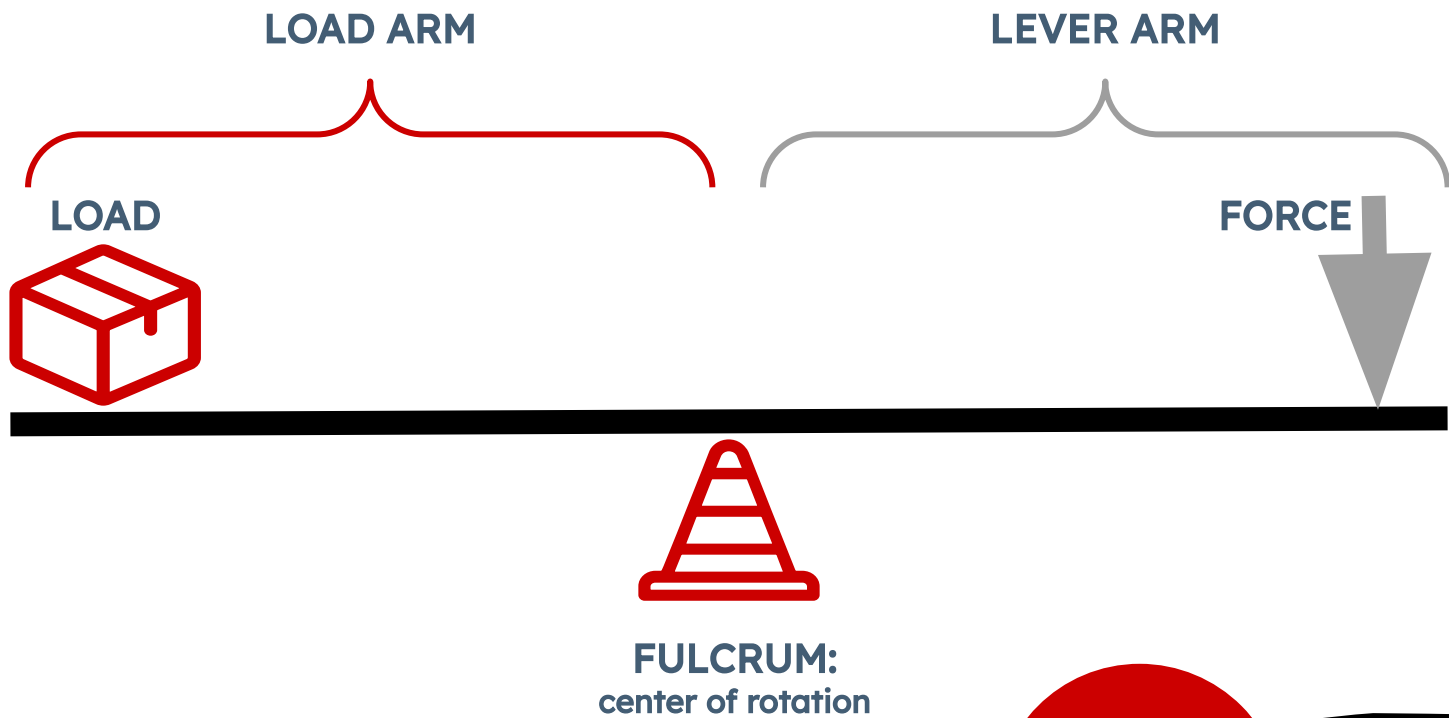
LEARNING OBJECTIVES

- 01** Identify and sketch the three classes of levers
- 02** Determine the mechanical advantage of a system
- 03** Apply the understanding of levers and mechanical advantage to joints in the body.



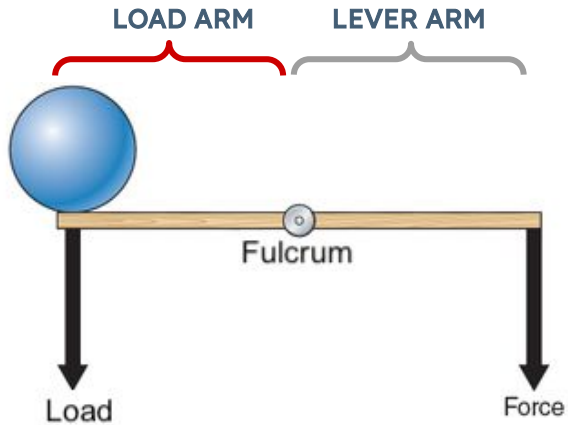
01 LEVERS





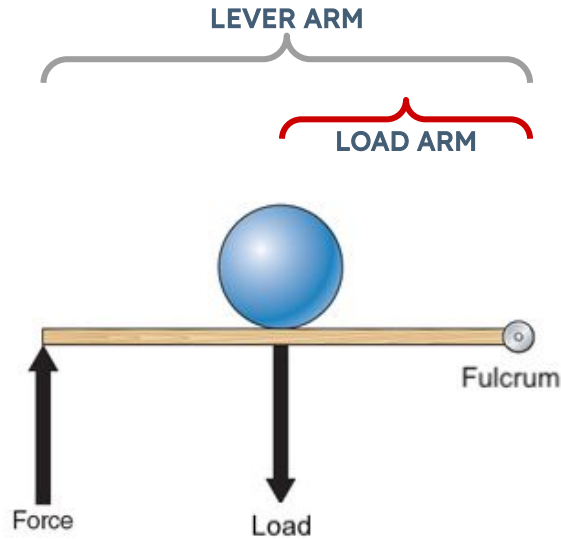
First Class Lever

Load, Fulcrum, Force



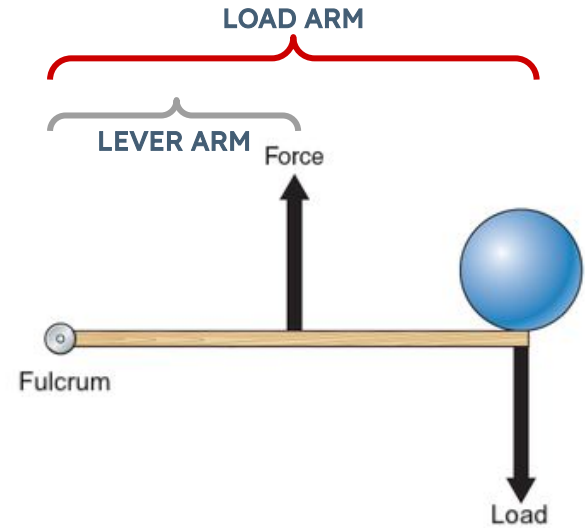
Second Class Lever

Force, Load, Fulcrum

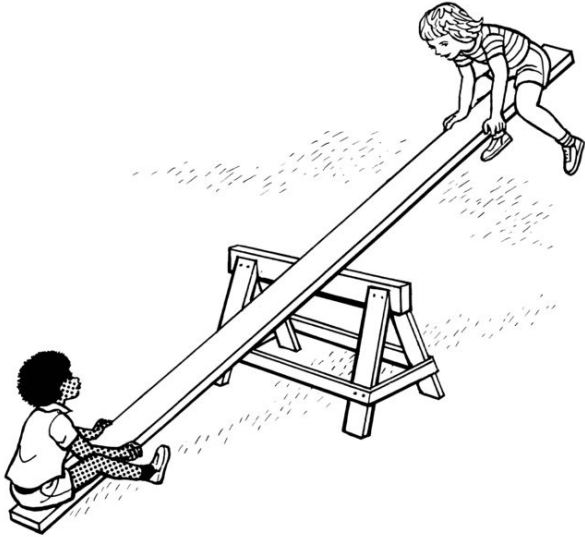


Third Class Lever

Fulcrum, Force, Load



First Class Lever



Second Class Lever

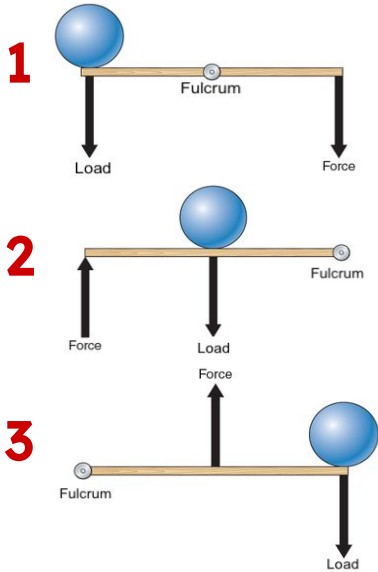


Third Class Lever

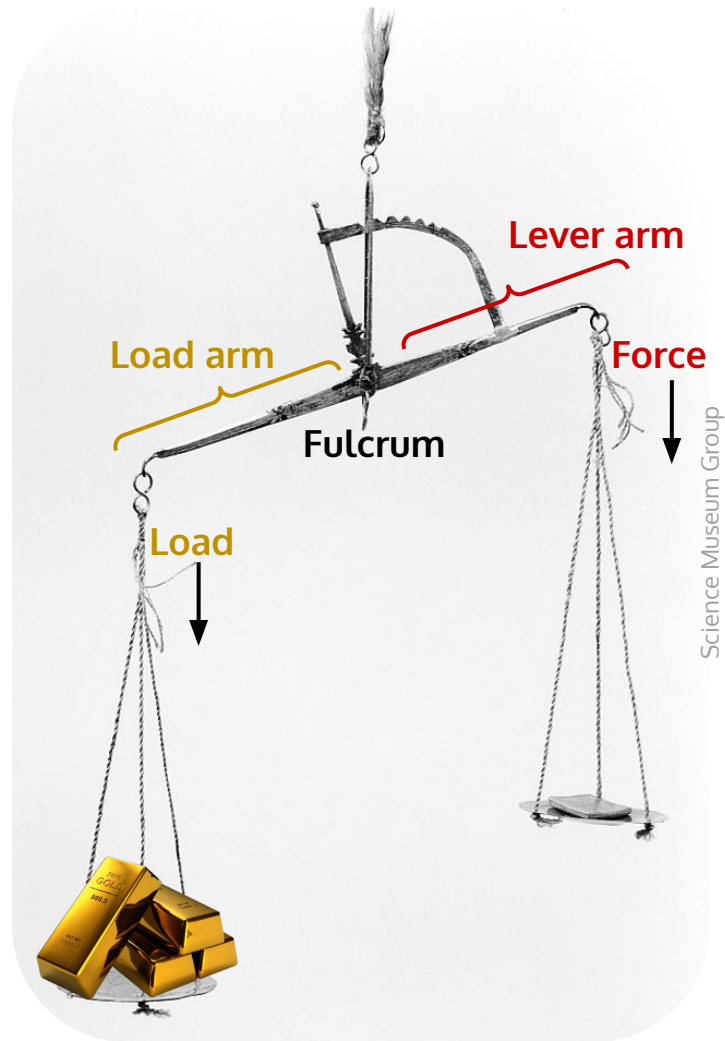


PRACTICE

Define the lever class and the load, fulcrum, force, load arm, and lever arm in the system.

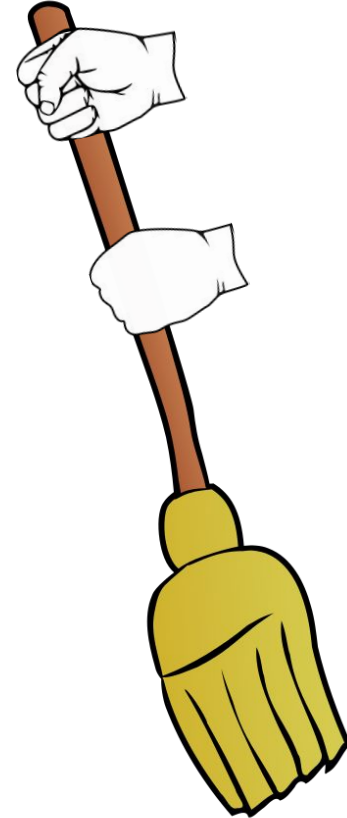
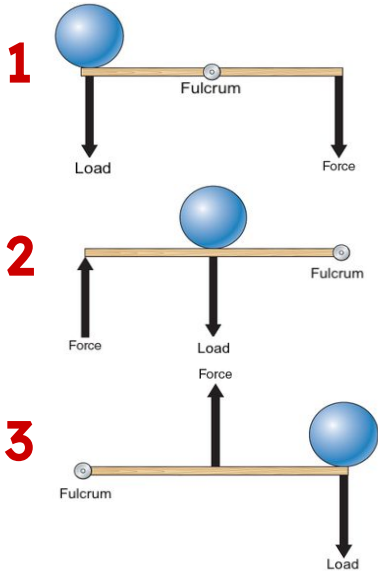


1st



PRACTICE

Define the lever class and the load, fulcrum, force, load arm, and lever arm in the system.



3rd

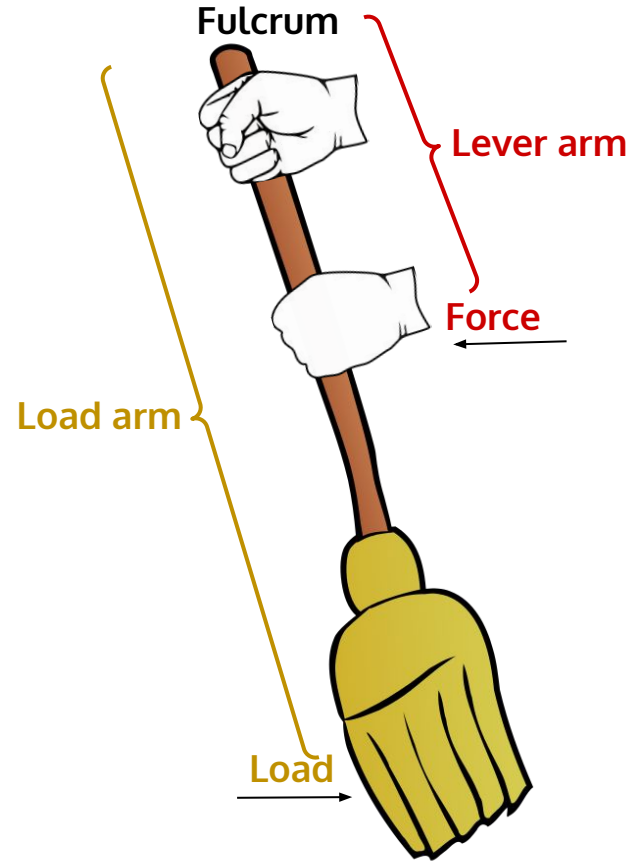
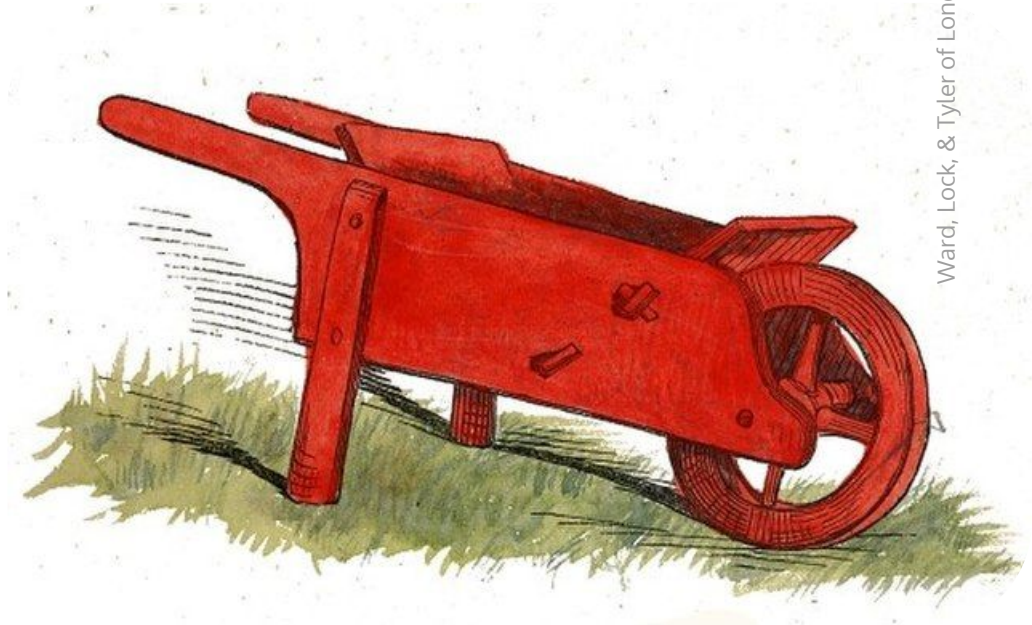
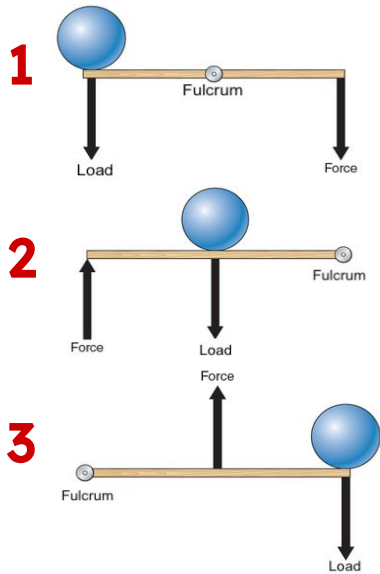


Photo courtesy of Wikimedia Commons

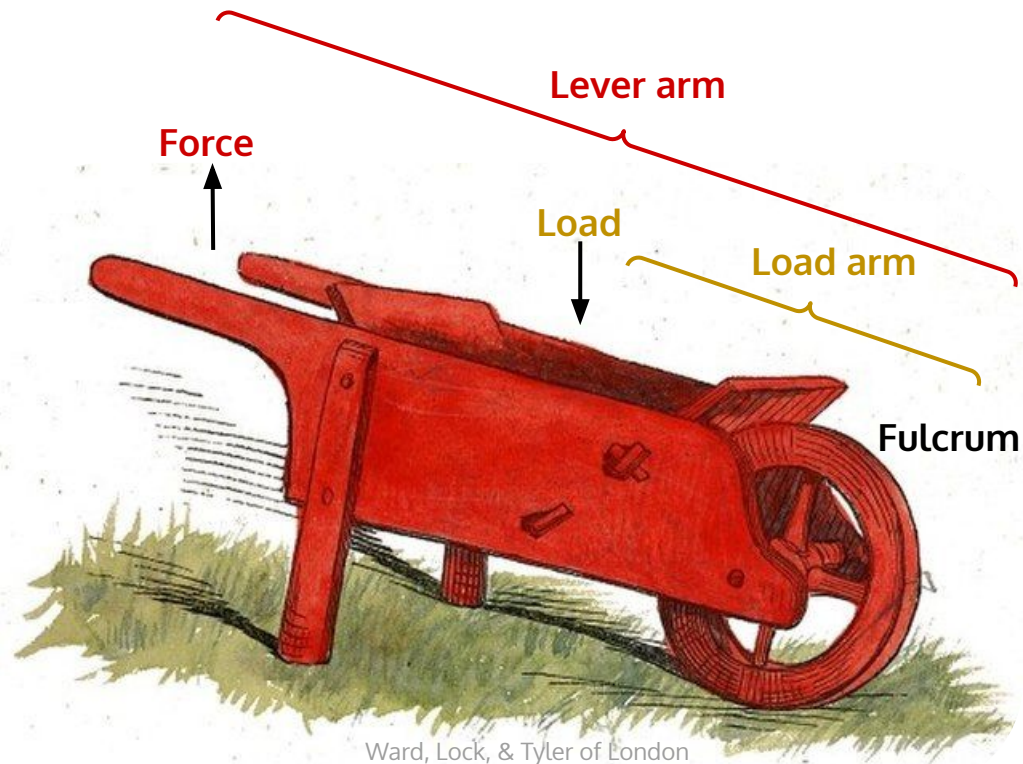
PRACTICE

Define the lever class and the load, fulcrum, force, load arm, and lever arm in the system.



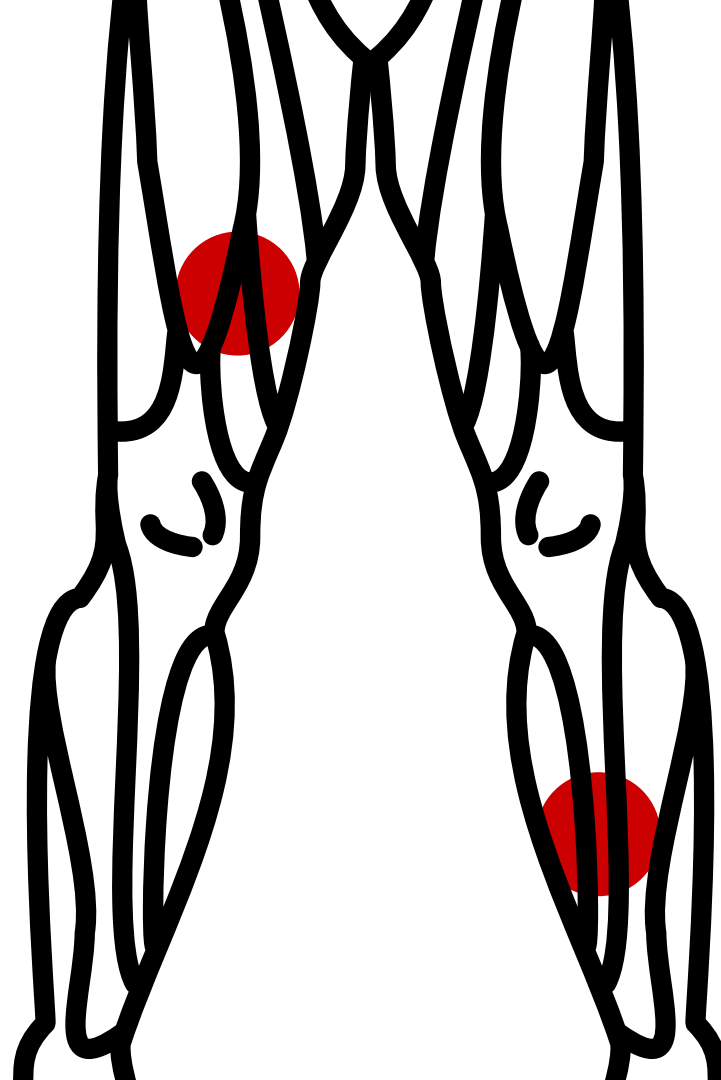
Ward, Lock, & Tyler of London

2nd



02

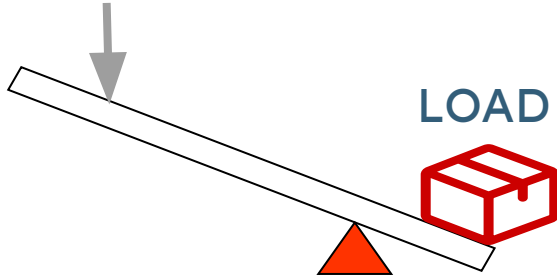
MECHANICAL ADVANTAGE



MECHANICAL ADVANTAGE:

Ratio of load weight : force applied

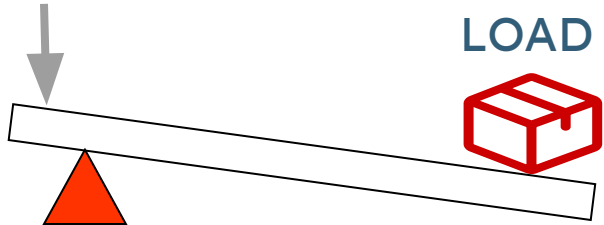
FORCE



Power system 1st & 2nd class

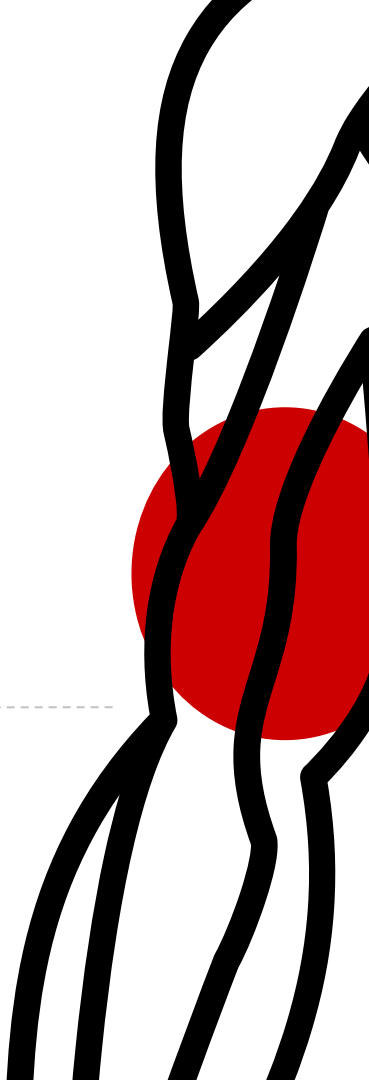
- Lever arm > load arm
- Higher mechanical advantage

FORCE

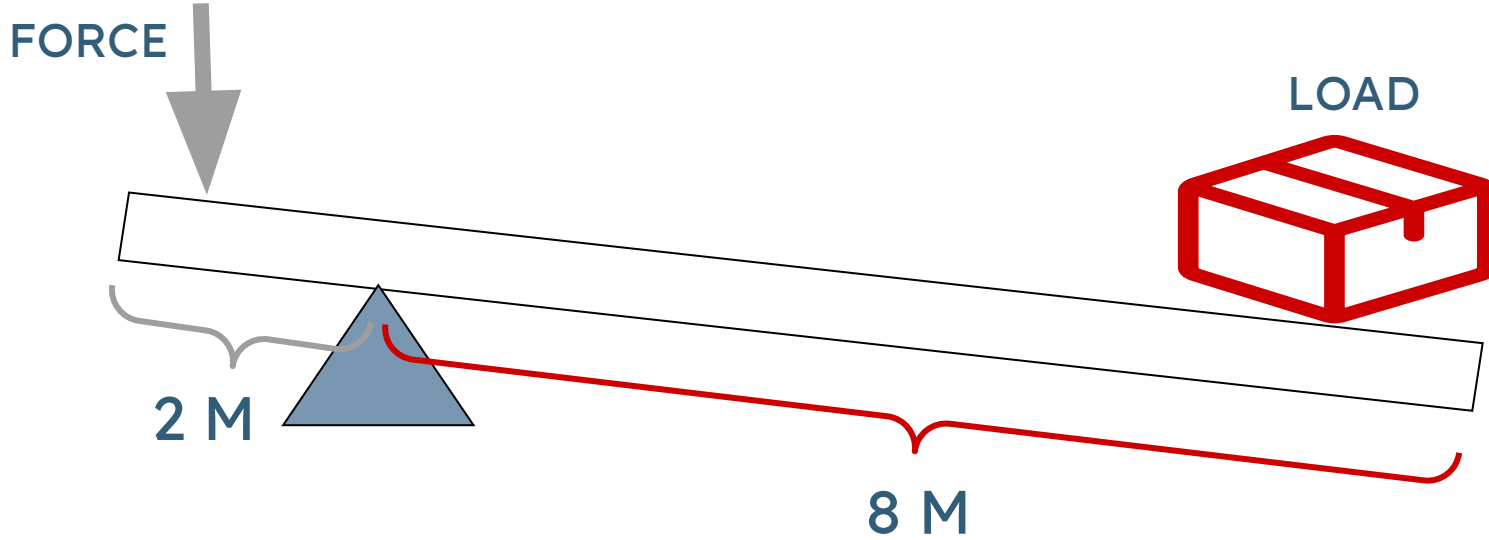


Speed system 3rd class

- Load arm > lever arm
- Lower mechanical advantage

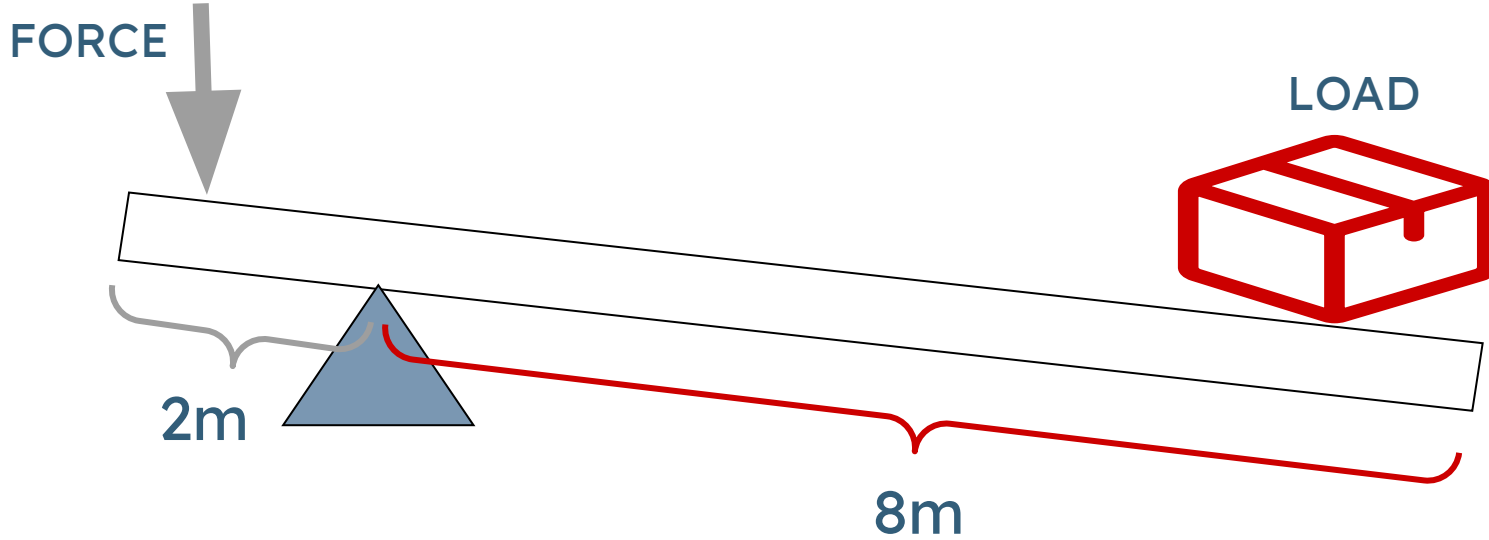


$$\text{Mechanical Advantage} = \frac{\text{Input Distance}}{\text{Output Distance}} = \frac{\text{Output Force}}{\text{Input Force}}$$



Is this a power or a speed system?

$$\text{Mechanical Advantage} = \frac{\text{Input Distance}}{\text{Output Distance}} = \frac{\text{Output Force}}{\text{Input Force}}$$



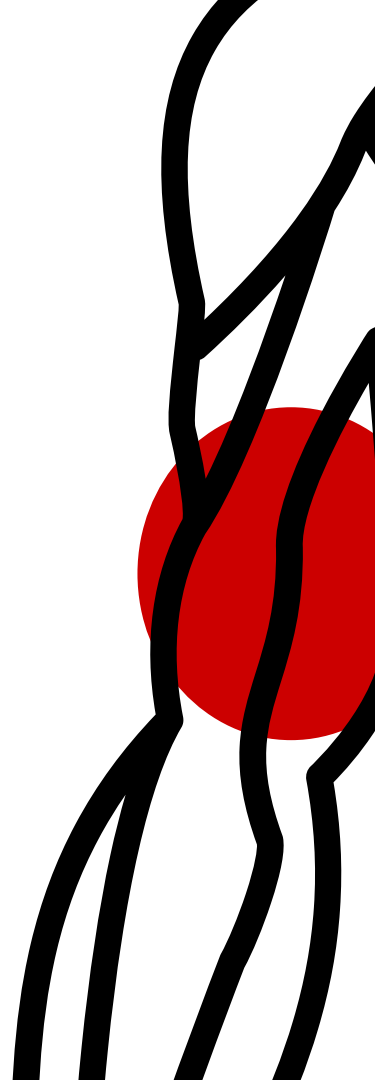
$$MA = 2m/8m = 1/4$$

$$MA < 1$$

Speed System

03

BIOMECHANICS

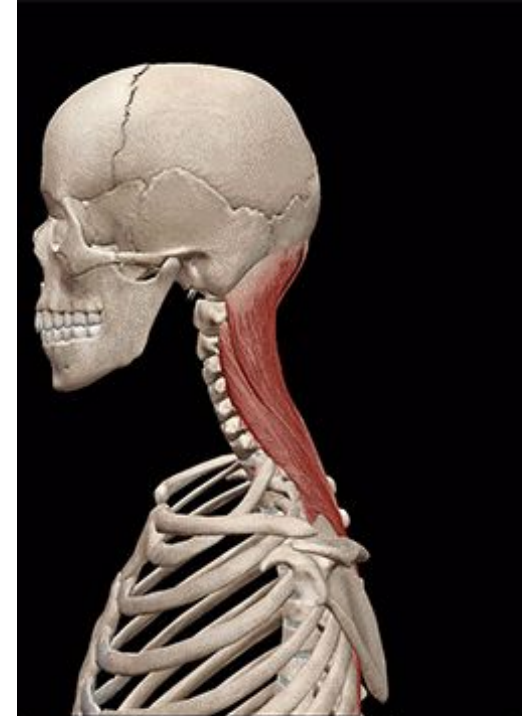
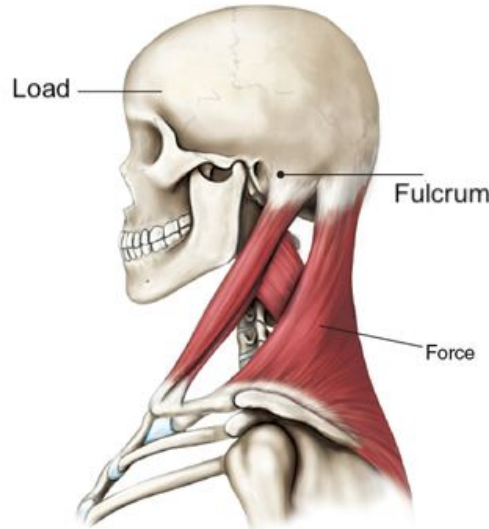
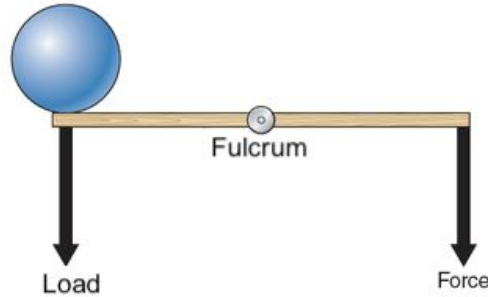


Biomechanics: First class lever

Load: Body weight or mass applied from an object

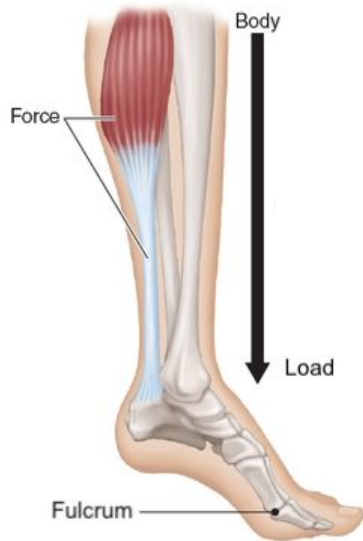
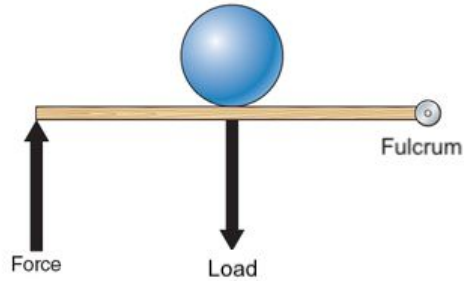
Force: Muscular force generated to move the system

Fulcrum: Joint

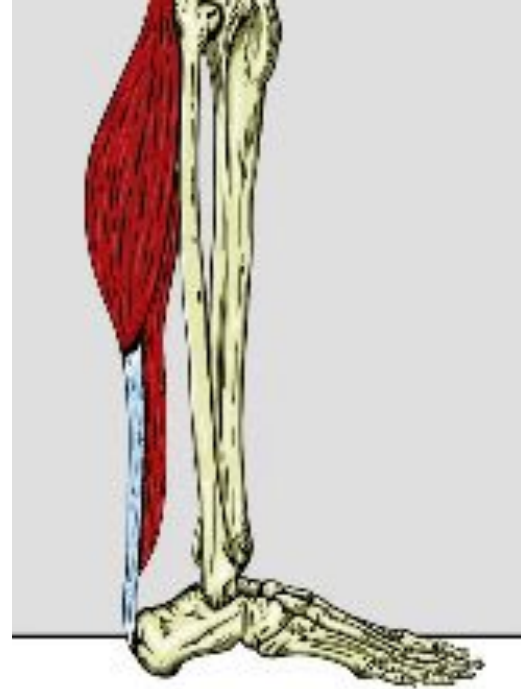


Behnke, 2001. Kinetic Anatomy

Biomechanics: **Second class lever**

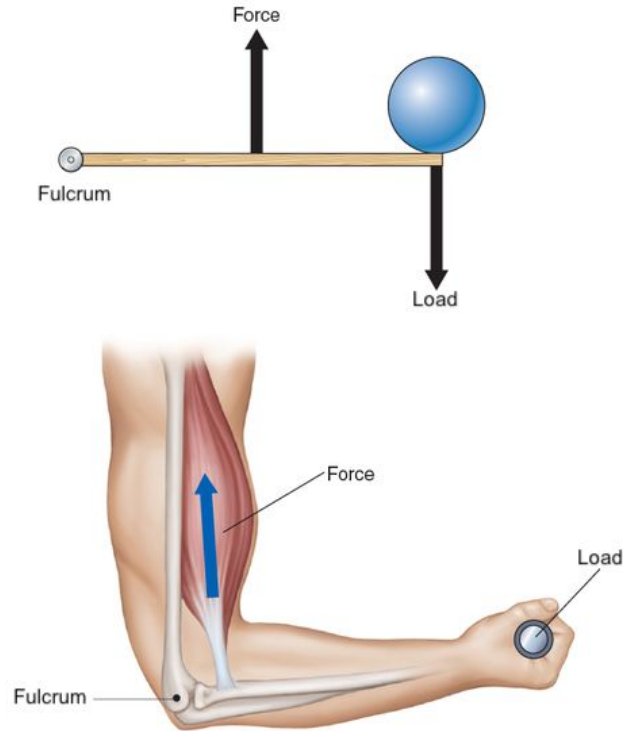


Behnke, 2001. Kinetic Anatomy

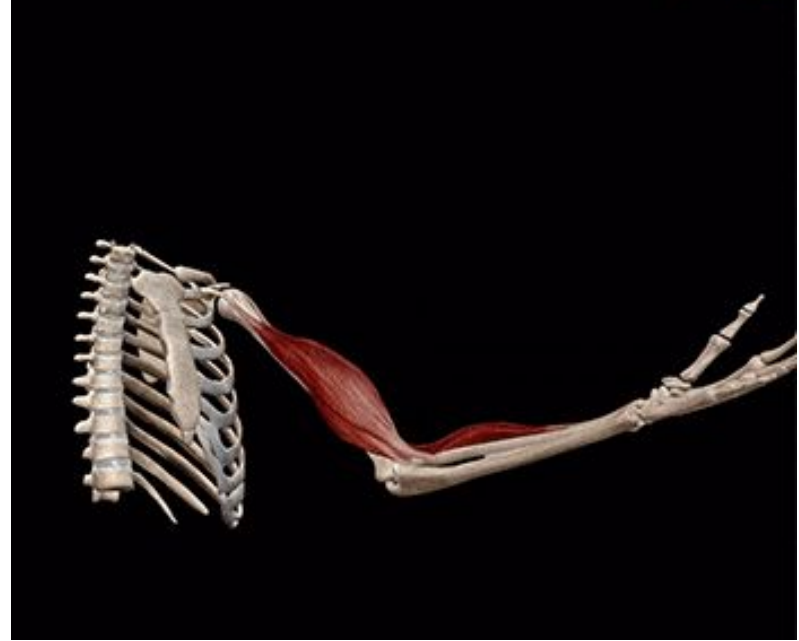


Courtesy of KinesologyKris.com

Biomechanics: **Third class lever**



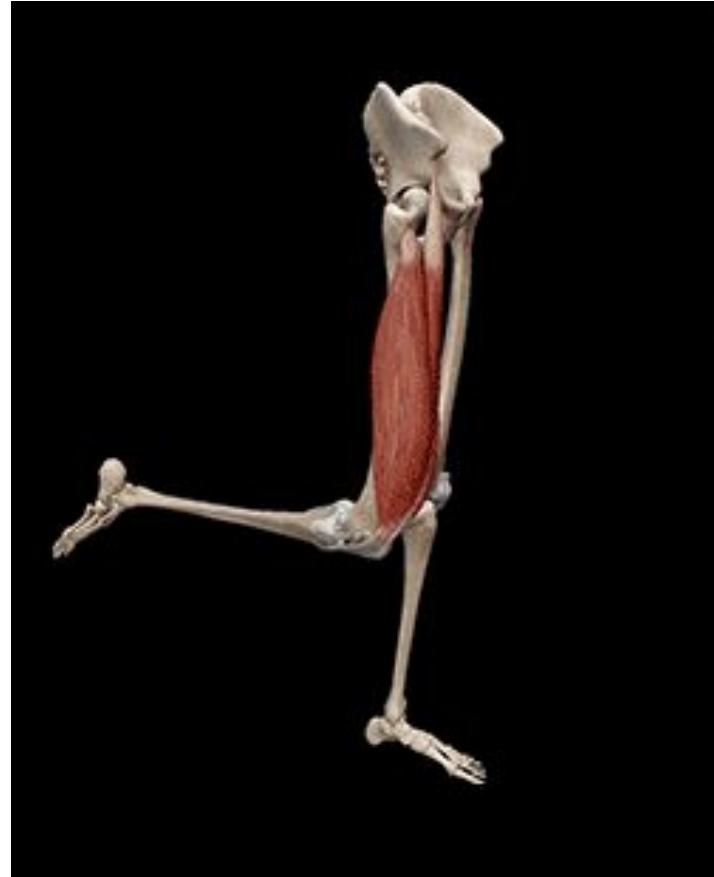
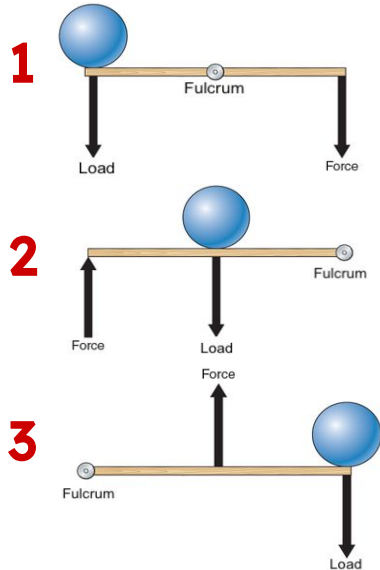
Behnke, 2001. Kinetic Anatomy



Muscles & Kinesiology

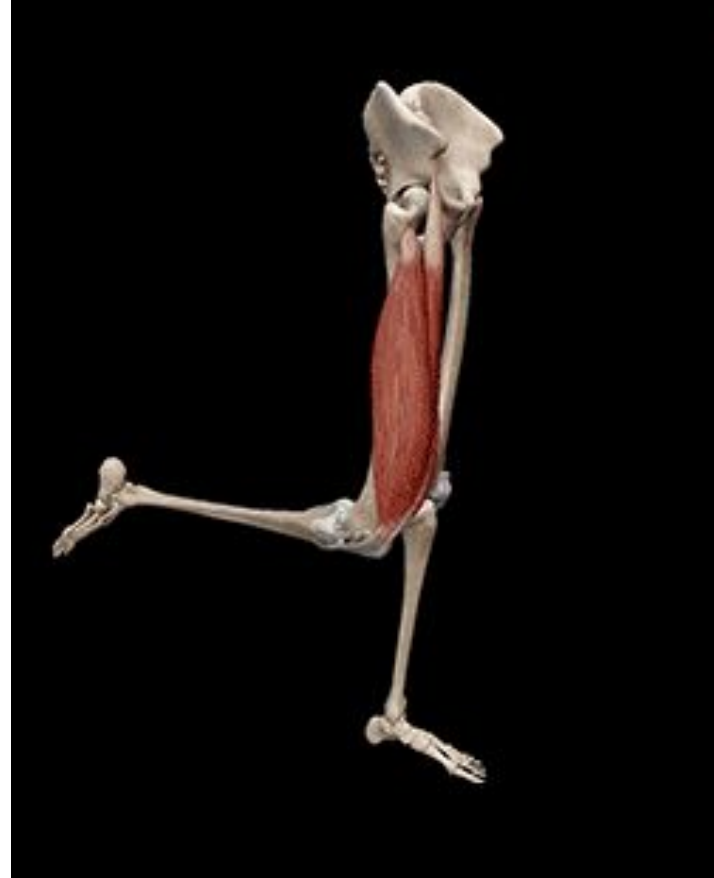
Your turn!

What lever system occurs during knee extension?
Define all the lever components



Your turn!

What lever system occurs during knee extension?



Mika & Goya are patiently waiting for the ball to be thrown. What lever class is present at the elbow joint when throwing the ball (Elbow extension)? Define the fulcrum, force, load, lever arm, and load arm of the system.

Hint Consider muscle attachment sites



I've decided to break out my ball launcher to play fetch. How does the addition of the launcher change the biomechanics of the system? Consider mechanical advantage.

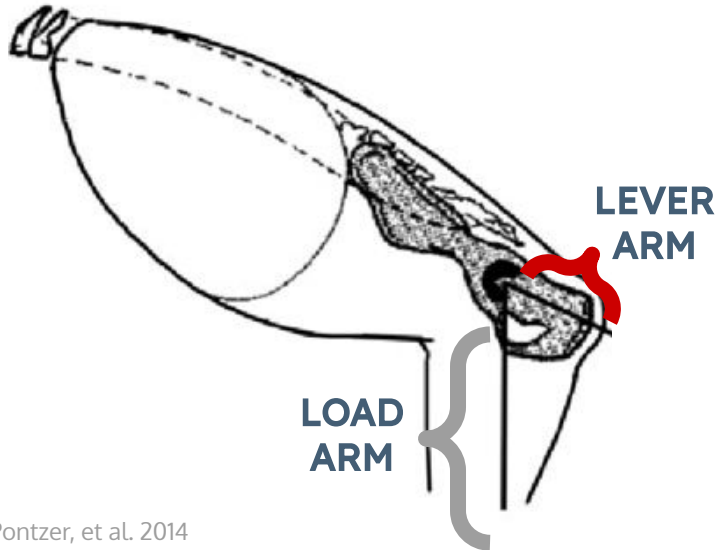


Comparative hip extension

Chimpanzee

- Long ischium
- Short hind limbs

Long lever arm =
POWER SYSTEM

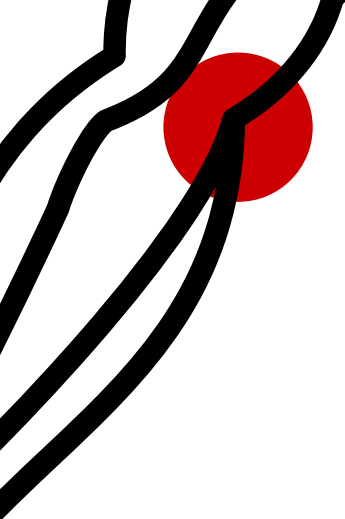


Human

- Short ischium
- Long lower limbs

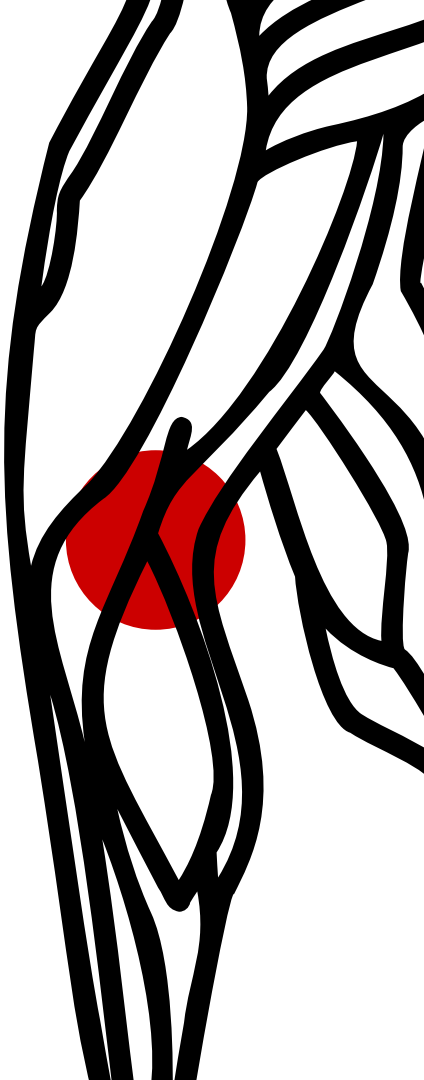
Long load arm =
SPEED SYSTEM





04 TUTORIAL

How to measure mechanical
advantage on MorphoSource





THANKS!

Questions/comments?
amg812@psu.edu

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